Devi Mutyala

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Education

University of Texas at Austin **2021 – 2026**
Ph.D. in Mechanical Engineering, Oden Institute (GPA: 3.95/4.0) *Austin, TX*

- **Publications:** 2× *ACS Nano* (2025) on Graphene & Boron Nitride for Hydrogen Fuel Cell applications (co-authored with Prof. Deji Akinwande). **Patent:** Venturi Air Cooler, IN202011018315 (2024).

Indian Institute of Technology, Delhi **2019**
M.S. (Research) in Mechanical Engineering (CGPA: 9.22/10) *New Delhi, India*

Indian Institute of Technology, Kanpur **2016**
B.Tech. in Aerospace Engineering (CGPA: 8.2/10) *Kanpur, India*

Technical Skills

ML & Deep Learning: PyTorch, TensorFlow, Hugging Face Transformers, Scikit-Learn, OpenCV, torchvision, BitsAndBytes (4-bit Quantization), ChromaDB (Vector DB)
Generative AI & LLMs: OpenAI API (GPT-4o, Whisper, TTS), Google Gemma, Meta Llama, Gemini, LangChain, LiteLLM, Ollama, LoRA, QLoRA, Fine-Tuning, Prompt Engineering
Agentic AI & RAG: Agentic AI, OpenAI Tool Calling (Function Calling), Multi-Agent Systems, RAG, Query Rewriting, LLM Reranking, LLM-as-a-Judge, Semantic Search
Programming & Databases: Python, SQL, SQLite, MongoDB, ReactJS, R, MATLAB, C
Tools & Infrastructure: Gradio, BeautifulSoup, Pyannote, Pandas, Requests, lxml, Docker, Git, AWS SageMaker, Jupyter

Professional Experience

Intel Corporation - Data Scientist (Internship) **May 2025 – Dec. 2025**
Design & Patterning AI Group (DPAI) *Santa Clara, CA*

- Engineered an end-to-end **Computer Vision (CV)** deep learning pipeline for **semiconductor chip defect detection**, spanning synthetic **layout-to-image generation**, configurable data augmentation, CNN model training, and evaluation; deployed as a modular component into Intel's production defect commonality detection codebase.
- Architected **positive-only supervised learning** workflows to tackle a **single-class, label-scarce** setting (~ 100 defect samples, **zero non-defect data**), enabling robust defect pattern modeling without requiring negative-class supervision.
- Designed and trained a **CNN-based Prototypical Part Network (ProtoPNet)** that discovers **local image patches** consistently recurring across defect samples but scarcely in randomly sampled layouts; optimized prototype coverage to **maximize activation on defect images** and **minimize false activation** on randomly sampled patches, producing **interpretable defect localizations (XAI)** directly on dataset images.
- Delivered a modular, production-ready **end-to-end ML pipeline** — data augmentation, model training, and results reporting — integrated into an existing codebase to enable automated, **explainable defect commonality analysis** at scale.

AI & Machine Learning Projects

Advanced Retrieval-Augmented Generation (RAG) System | *OpenAI SDK, LiteLLM, LangChain, ChromaDB, Gradio, Tenacity*

- Architected a dual-pipeline RAG system benchmarking a standard **LangChain** baseline against a custom advanced retrieval engine, improving **MRR from 0.85 to 0.88** and boosting answer accuracy to **4.51/5** across 150 benchmark tests.
- Engineered a multi-stage ingestion pipeline using LLM-driven **semantic document segmentation** and zero-shot chunk summarization to maximize embedding contextual density within a persistent **ChromaDB** vector store.
- Implemented dynamic **query rewriting**, concurrent multi-query retrieval, and **LLM-based zero-shot reranking**, achieving **95.5% keyword coverage** — a 3.5-point gain over the 92% baseline — validated via a custom **LLM-as-a-judge** evaluation framework with interactive Gradio dashboards.

Medical Visual Question Answering (VQA) | *Gemma, QLoRA, Hugging Face, Gemini, BioCLIP, DenseNet, BERT*

- Fine-tuned **Gemma-3-4B-it** via **QLoRA** on the VQA-RAD radiology dataset to generate detailed clinical explanations for radiology scans, achieving **BLEU 0.23** and **ROUGE-1 F1: 0.48** on a custom generative benchmark.
- Engineered a synthetic data augmentation pipeline using **Gemini-2.0-Flash** to convert sparse short-answer training data into rich descriptive examples, improving training convergence without additional human annotation.
- Developed two classification baseline pipelines (**BioCLIP + BERT** and **DenseNet121 + BiLSTM**) to validate the fine-tuning approach against strong discriminative architectures on binary and closed-ended VQA tasks.

Multimodal Flight Assistant | *Agentic AI, OpenAI Tool Calling, Whisper, PyTorch, SQL*

- Architected a **multi-agent system** with a **dual-LLM orchestration architecture** — one model for conversational dialogue, one for geographic entity normalization — enabling robust natural-language flight booking workflows.
- Engineered an **Agentic AI** pipeline leveraging **OpenAI Function Calling** to securely map user intent to real-time **SQL** queries, enabling autonomous ticket price retrieval and role-based authenticated price updates.
- Built an end-to-end multimodal interface integrating **OpenAI Whisper (STT)**, **TTS**, and **Gradio**, delivering fully hands-free voice-to-voice customer interactions.

Smart Audio Scribe | *OpenAI Whisper, Pyannote, Meta Llama, Hugging Face, PyTorch, BitsAndBytes*

- Architected a speech processing pipeline combining **OpenAI Whisper** and **Pyannote** speaker diarization to produce structured, speaker-attributed transcripts from multi-speaker audio recordings.
- Engineered a privacy-first **Generative AI** engine running **4-bit quantized** local **Meta Llama** models (1B/3B/8B-Instruct) to autonomously classify audio content type (Meeting / Speech / Conversation) and synthesize **context-aware meeting minutes** and action items without external API calls.

AI Financial News Engine | *GPT-5-nano, GPT-4.1-mini, BeautifulSoup, Python, Requests*

- Engineered an intelligent financial data pipeline using **GPT-5-nano** to autonomously discover, filter, and rank up to 10 maximally relevant news links from Yahoo Finance, eliminating manual curation.
- Implemented a summarization layer using **GPT-4.1-mini** to synthesize multi-source financial content into structured markdown **stock news summaries** with robust error handling across high-volume scraping runs.

Find Common Object (Explainable AI / ProtoPNet) | *PyTorch, torchvision, CNN, Matplotlib*

- Developed a **CNN**-based **Explainable AI** model implementing **Prototypical Part Networks (ProtoPNet)** to automatically discover and localize common sub-objects across image classes.
- Designed models learning human-interpretable image structural prototypes rather than black-box embeddings, providing spatial localization through **bounding boxes** and **heatmaps**.

Engineering & Research Experience

Oden Institute, University of Texas at Austin – Graduate Researcher **Sept. 2022 – May 2025**

Advisor: Prof. Narayana Aluru *Austin, TX*

- Investigating complex physics of nanoscale systems to explore the development of a **nanofluidic memristor**.
- Employing **Molecular Dynamics simulations (LAMMPS)** and **Poisson-Nernst-Planck continuum models (COMSOL)** to analyze ionic transport and electrokinetic behavior in nanoscale environments.
- Utilizing **Python** to develop analytical frameworks and conduct comprehensive **data analysis**, uncovering insights connecting physical mechanisms with anomalies in large-scale simulation results.

Indian Institute of Technology, Madras – Project Associate **Aug. 2020 – Apr. 2021**

Department of Aerospace Engineering *Chennai, India*

- Executed **Large-Eddy Simulations (LES)** of high Reynolds number channel flows with non-equilibrium effects.
- Implemented **wall-modeled LES (wall stress model)** for non-equilibrium flows and sensor-based computations for accurate transition flow modeling into high-fidelity **in-house CFD solvers**.

Havells India Ltd. – Research Engineer **July 2019 – Aug. 2020**

Center for Research and Innovation *Delhi, India*

- **Patented Venturi Air Cooler:** Designed an energy-efficient device eliminating water pumps by harnessing pressure/temperature profiles. Secondary inventor on patent **IN202011018315**.
- Developed analytical models leveraging **Blade Element theory** and **CFD** to design aerodynamic hardware, improving volumetric flow rates by 10% and **minimizing power consumption** by 3% for scalable production.

University of Texas at Austin / IIT Delhi – Teaching Assistant **2017 – 2025**

Mechanical Engineering Departments *Austin, TX & Delhi, India*

- Instructed tutorial sessions and conducted automated evaluations using Python for Applied Thermodynamics (2025), Computational Fluid Dynamics (2022), Propulsion (2021), and Gas Dynamics (2018).

Awards & Achievements

Academic Awards: Recipient of the **UT Austin Engineering Fellowship** (2021-2025) and the **IIT Kanpur Academic Excellence Award** awarded to the top 5% of the class (2014-2015).

Leadership and Service: Recipient of the **Institute Blues** (2016), the highest honor at IIT Kanpur for exceptional sports leadership and institutional impact as Team Captain and Deputy Contingent Leader.

Athletics Achievements: Attained 8 Inter IIT Athletics records and 11 Gold medals in track and field at IIT. Served as Captain of the Institute Athletics Team and Deputy Contingent Leader.